Software Requirement Specifications

The Spare Grid

An online Spare part dealership

Submitted by

Muhammad Affan (S23BINFT1E02067)

Submitted to

Dr. Rizwan Ali Shah

Department of Information Technology

Faculty of Computing

The Islamia University of Bahawalpur

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Introduction

1. Purpose of this Document

- The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive and detailed description of the requirements for the development of a **The Spare Grid.**
- This document is intended to serve as a formal agreement and reference guide between the stakeholders, development team, and quality assurance team throughout the entire software development lifecycle.
- Specifically, this SRS aims to:
- Define the functional and non-functional requirements of a web-based marketplace that enables **motor spare parts dealers from multiple countries** to register, manage, and promote their businesses online.
- Describe how **international customers** can browse, search, and place orders for motor spare parts from verified dealers across the globe.
- Establish a clear understanding of the system's expected behavior, including **multi-country dealer registration**, **product listing management**, **international order processing**, and **cross-border shipping integration**.
- Serve as the foundation for system design, development, testing, and future maintenance of the platform.
- Eliminate ambiguity between stakeholders and the development team by documenting all agreed-upon requirements in a single, structured reference.
- This document is intended to be used by:
- **Project Managers** – for planning, scope management, and delivery tracking
- **System Architects & Developers** – as a technical blueprint for building the platform
- **QA/Test Engineers** – for creating test cases and validating system behavior
- **Business Stakeholders & Clients** – to verify that the system meets business objectives

- **UI/UX Designers** – to understand user workflows and interface requirements

2. Scope

2.1 Product Name

Global Motor Spare Parts Marketplace (or your chosen product name)

2.2 Product Overview

The Spare Grid is a web-based platform designed to bridge the gap between **motor spare parts dealers worldwide** and **customers seeking automotive components globally**. The platform will provide a centralized digital environment where dealers can register their businesses, list their products, and manage orders, while customers can search, compare, and purchase spare parts from verified international suppliers.

2.3 What the System Will Do

The platform will provide the following core capabilities:

For Dealers:

- Allow motor spare parts dealers from different countries to register and create verified business profiles
- Enable dealers to list, manage, and update their spare parts inventory with detailed product information

- Allow dealers to set pricing in their local currency with automatic conversion support
- Provide dealers with an order management dashboard to process and track incoming orders
- Support dealers with shipping and logistics integration for cross-border deliveries
- Provide sales analytics and reporting tools for business insights.

For Customers:

- Allow customers from any country to browse and search for specific motor spare parts
- Enable customers to filter products by category, vehicle type, brand, country of dealer, price range, and availability
- Allow customers to place international orders.
- Provide customers with real-time order tracking and delivery status updates
- Allow customers to review and rate dealers and products

For Platform Admin:

- Provide an admin panel to manage dealer registrations, verifications, and approvals
- Monitor transactions, disputes, and platform activity
- Manage platform-wide settings such as supported countries.

2.4 What the System Will Not Do

To clearly define boundaries, the following are outside the scope of this platform:

- The platform will **not** manufacture or physically stock any spare parts
- The platform will **not** provide vehicle repair or mechanical services
- The platform will **not** handle customs clearance directly (dealers and customers are responsible for their respective country's import/export regulations)
- The platform will **not** support offline or desktop application versions in this phase

- The platform will **not** include a built-in logistics/courier service (third-party integrations will be used)

2.5 Target Users

User Type	Description
Dealers	Motor spare parts businesses from multiple countries seeking global reach
Customers	Individual buyers or garages worldwide looking for specific spare parts
Platform Admin	Internal team managing platform operations, verifications, and disputes

2.6 Benefits of the System

- Provides dealers with access to a **global customer base** beyond their local markets
- Gives customers access to a **wider variety of spare parts** that may not be available locally
- Reduces the complexity of **international B2C trade** in the automotive parts industry
- Creates a **trusted, verified marketplace** reducing the risk of fraud for both buyers and sellers

2.7 Platform Boundaries

Boundary	Details
Geography	Accessible globally, supporting dealers and customers from all countries
Device Support	Web browsers on desktop and mobile (responsive design)
Language	English as primary language (multilingual support in future phases)

Currency	Multi-currency support with real-time exchange rate integration

3. Overview

3.1 Product Perspective

The **Spare Grid** is a newly developed, independent web-based platform. It is not a replacement or extension of any existing system but rather a standalone solution designed to serve the global automotive spare parts industry.

The platform will operate as a **multi-vendor, multi-country e-commerce marketplace** specifically tailored for motor spare parts. It will interact with several external systems and services to deliver its full functionality, including:

- **Shipping & Logistics APIs** (e.g., DHL, FedEx, ShipStation) for cross-border delivery management
- **Email & Notification Services** (e.g., SendGrid, Twilio) for alerts and communications
- **Cloud Hosting Infrastructure** for scalability and global availability
- **Google Maps / Geolocation API** for dealer location display and regional filtering

The system will be accessible through standard web browsers on both desktop and mobile devices without requiring any additional software installation.

3.2 Product Functions

At a high level, the platform will perform the following major functions:

Dealer Side:

- Business registration and profile creation with country-specific details
- Product/spare parts listing with categories, vehicle compatibility, images, and pricing
- Inventory management and stock tracking

- Order receiving, processing, and status updating
- Earnings dashboard and withdrawal management
- Communication with customers through messaging system

Customer Side:

- Account creation and profile management
- Advanced search and filtering of spare parts by make, model, year, category, brand, and country
- Product comparison and wishlist management
- Shopping cart and international checkout process
- Multi-currency payment processing
- Order placement, tracking, and history management
- Dealer and product review and rating system

Admin Side:

- Dealer registration review and approval/rejection
- User and dealer account management
- Product listing moderation and quality control
- Transaction monitoring and dispute resolution
- Platform configuration and settings management
- Reports and analytics on platform performance, sales, and user activity
- Commission and fee management

3.3 User Classes and Characteristics

User Class	Description	Technical Level	Access Level
Dealer	Motor spare parts business owners or representatives registering from various countries	Basic to Intermediate	Dealer Dashboard
Customer	Individual buyers, garage owners, or businesses purchasing spare parts globally	Basic	Customer Portal
Platform Administrator	Internal staff managing and operating the platform	Advanced	Full Admin Panel

Guest User	Unregistered visitors browsing the platform without an account	Basic	Browse Only

3.4 Operating Environment

The platform will operate in the following environment:

Server Side:

- Cloud-based hosting environment (e.g., AWS, Google Cloud, or Azure)
- Linux-based web server (e.g., Apache or Nginx)
- Relational database management system (e.g., MySQL or PostgreSQL)
- Backend developed using a modern framework (boot strap)

Client Side:

- Compatible with all modern web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari
- Responsive design supporting desktop, tablet, and mobile screen sizes
- No additional plugins or software required for end users

Network:

- Requires a stable internet connection
- All data transmission secured via HTTPS/SSL encryption

3.5 Design and Implementation Constraints

The following constraints will influence the design and development of the platform:

- The platform must comply with **international data privacy regulations** such as GDPR (Europe) and other regional data protection laws
- The system must support **multiple currencies and tax structures** varying by country
- The platform must handle **cross-border trade regulations** and provide necessary documentation support (invoices, receipts)

- The system must be designed for **high availability** with minimal downtime given its global user base across different time zones
- **Language and cultural differences** must be considered in the UI/UX design for international usability
- The platform must be **scalable** to accommodate a growing number of dealers, products, and customers without performance degradation

3.6 Assumptions and Dependencies

Assumptions:

- Dealers are assumed to have a legitimate registered business and will provide valid business documentation during registration
- Customers are assumed to have basic knowledge of the spare parts they are searching for (part name, vehicle make/model)
- All dealers are responsible for the accuracy of their product listings, pricing, and availability
- Dealers and customers are responsible for complying with their respective country's import/export laws
- A stable internet connection is available for all platform users

Dependencies:

- The platform depends on **third-party payment gateway APIs** for transaction processing
- Shipping cost estimation and tracking depend on integration with **third-party logistics providers**
- Email and SMS notifications depend on **external communication service providers**
- Platform availability depends on the uptime and reliability of the chosen **cloud hosting provider**

3.7 System Context Diagram (Description)

The platform sits at the center of interactions between the following entities:

- **Dealers** → Register, list products, manage orders via Dealer Dashboard
- **Customers** → Browse, search, order, and track via Customer Portal
- **Admin** → Monitor, approve, and manage via Admin Panel
- **Logistics API** → Manages shipping rates, labels, and tracking
- **Email/SMS Service** → Sends notifications and alerts to all users
- **Cloud Server** → Hosts and serves the entire platform globally

4. General Description

4.1 System Overview

The Spare Grid is a comprehensive, web-based multi-vendor platform built to serve the international automotive spare parts industry. The system creates a digital bridge between **motor spare parts dealers operating across different countries** and **customers worldwide** who need reliable access to automotive components.

The platform functions as a **trusted global marketplace** where:

- Dealers can digitally establish their business presence, showcase their inventory, and fulfill international orders
- Customers can conveniently search, compare, and purchase genuine spare parts from verified dealers around the world
- Platform administrators can oversee operations, ensure quality, and maintain a secure and fair trading environment

The system is designed to eliminate geographical barriers in the automotive parts supply chain, making it easier for customers to find rare or region-specific parts and for dealers to expand their market reach beyond local boundaries.

4.2 System Architecture Overview

The platform will be built on a **three-tier architecture**:

Tier	Description
Presentation Layer	The front-end interface accessible by dealers, customers, and admins through web browsers
Application Layer	The back-end business logic handling registrations, orders, payments, and communications
Data Layer	The database storing all user profiles, product listings, transactions, and platform data

The system will follow a **client-server model** hosted on a cloud infrastructure, ensuring global accessibility, scalability, and high availability across different time zones.

4.3 Key Modules of the System

The platform is divided into the following major modules:

4.3.1 Dealer Management Module

- Dealer registration with business verification
- Business profile creation and management
- Document upload for business authentication
- Dealer dashboard for managing products and orders
- Performance analytics and sales reports

4.3.2 Customer Management Module

- Customer account registration and login
- Profile management and address book
- Order history and tracking
- Wishlist and saved searches
- Review and rating management

4.3.3 Product & Inventory Management Module

- Spare parts listing with detailed specifications
- Vehicle compatibility mapping (make, model, year)
- Category and subcategory management
- Stock level tracking and low stock alerts
- Product image and document uploads

4.3.4 Search & Discovery Module

- Advanced search engine for spare parts
- Filters by category, brand, vehicle type, country, price, and availability
- Search suggestions and autocomplete
- Featured and sponsored product listings
- Related products and recommendations

4.3.5 Order Management Module

- Shopping cart and checkout process
- International order placement and confirmation
- Order status tracking (placed, confirmed, shipped, delivered)
- Order cancellation and return request handling
- Invoice and receipt generation

4.3.6 Payment Module

- Multi-currency payment support
- Integration with international payment gateways (Stripe, PayPal, etc.)
- Secure transaction processing (PCI-DSS compliant)
- Commission deduction and dealer payout management
- Refund and dispute handling

4.3.7 Shipping & Logistics Module

- Shipping cost estimation based on weight, dimensions, and destination
- Integration with international courier services (DHL, FedEx, etc.)
- Shipment tracking and delivery status updates
- Multiple shipping options (standard, express, economy)
- Customs documentation support

4.3.8 Communication Module

- In-platform messaging between dealers and customers
- Automated email and SMS notifications for orders, payments, and shipping
- Dealer inquiry and quotation request system
- Platform announcements and alerts

4.3.9 Review & Rating Module

- Customer reviews and star ratings for dealers
- Customer reviews and ratings for individual products
- Dealer response to reviews
- Admin moderation of reviews

4.3.10 Admin Control Module

- Dealer registration approval and rejection
- User account management and suspension
- Product listing moderation
- Transaction and dispute monitoring
- Commission and fee configuration
- Platform-wide reporting and analytics

4.4 User Interaction Overview

The following describes how each type of user interacts with the system:

4.4.1 Guest User Journey

```
Visit Platform → Browse Products → Search Spare Parts
→ View Product Details → View Dealer Profile
→ Prompted to Register/Login to Place Order
```

4.4.2 Customer Journey

```
Register/Login → Search for Spare Part → Filter Results
→ Compare Products → Add to Cart → Checkout
→ Select Shipping Method → Make Payment
→ Track Order → Receive Product → Leave Review
```

4.4.3 Dealer Journey

```
Register Business → Submit Verification Documents
→ Await Admin Approval → Set Up Profile
→ List Products → Receive Orders → Process & Ship Orders
→ Manage Payments → View Analytics
```

4.4.4 Admin Journey

```
Login to Admin Panel → Review Dealer Applications
→ Approve/Reject Registrations → Monitor Listings
→ Handle Disputes → Manage Commissions
→ Generate Reports → Configure Platform Settings
```

4.5 Data Flow Overview

The following describes the high-level flow of data within the system:

Flow	Description
Dealer → Platform	Business details, product listings, inventory updates, order status updates
Customer → Platform	Search queries, order requests, payment information, reviews
Platform → Dealer	Order notifications, customer inquiries, payment settlements
Platform → Customer	Search results, order confirmations, shipping updates, invoices
Platform ↔ Payment Gateway	Transaction requests, payment confirmations, refund processing
Platform ↔ Logistics API	Shipping rate requests, tracking number generation, delivery updates
Platform ↔ Currency API	Real-time exchange rate fetching for multi-currency display

4.6 General Constraints

The following general constraints apply to the overall system:

- The platform must be **available 24/7** to accommodate users across all global time zones
- All user data must be stored and processed in compliance with **international data privacy laws** including GDPR
- The system must support a **minimum of 10,000 concurrent users** without performance degradation
- All financial transactions must be encrypted and processed in compliance with **PCI-DSS standards**

- The platform must support **multiple languages** in future phases to serve a global audience
- The system must provide **role-based access control** ensuring each user type only accesses permitted features
- The platform must be **responsive and mobile-friendly** for users accessing via smartphones and tablets

4.7 General Assumptions

- All dealers registering on the platform are assumed to be **legally operating businesses**
- Customers are responsible for ensuring the spare parts they order are **compatible with their vehicles**
- The platform assumes that **third-party services** (payment gateways, logistics APIs) will maintain standard uptime and availability
- It is assumed that dealers will **accurately represent** their products, stock levels, and pricing
- The platform assumes that dealers and customers will comply with their respective country's **trade and import/export laws**

4.8 Future Scope (Out of Current Phase)

The following features are acknowledged but will be considered for future development phases:

- **Mobile Application** (iOS and Android) for dealers and customers
- **AI-powered part recommendation engine** based on vehicle type and history
- **Multilingual support** for non-English speaking markets
- **Augmented Reality (AR)** for part visualization and fitment checking
- **B2B bulk ordering system** for garages and automotive workshops
- **Integrated customs clearance assistance** for cross-border shipments
- **Dealer subscription plans** with premium listing features

5. Functional Requirements

5.1 User Registration & Authentication Module

5.1.1 Guest User

ID	Requirement
FR-001	The system shall allow guest users to browse and view product listings without registration
FR-002	The system shall allow guest users to search for spare parts using keywords and filters
FR-003	The system shall prompt guest users to register or login when attempting to place an order
FR-004	The system shall display dealer profiles and product details to guest users

5.1.2 Customer Registration & Login

ID	Requirement
FR-005	The system shall allow customers to register using email address and password
FR-006	The system shall allow customers to register using Google or Facebook OAuth
FR-007	The system shall send an email verification link upon customer registration
FR-008	The system shall allow customers to login using registered email and password
FR-009	The system shall provide a forgot password and password reset functionality via email
FR-010	The system shall allow customers to update their profile information including name, address, and contact number
FR-011	The system shall allow customers to manage multiple shipping addresses
FR-012	The system shall allow customers to deactivate their account

5.1.3 Dealer Registration & Login

ID	Requirement
FR-013	The system shall allow dealers to register by providing business name, country, contact details, and business type
FR-014	The system shall require dealers to upload valid business registration documents during registration
FR-015	The system shall allow dealers to specify the countries they can ship to during registration
FR-016	The system shall send a registration confirmation email to the dealer upon submission
FR-017	The system shall place dealer accounts in a **pending state** until reviewed and approved by admin
FR-018	The system shall notify dealers via email upon approval or rejection of their registration
FR-019	The system shall allow approved dealers to login using email and password

FR-020	The system shall allow dealers to update their business profile, contact information, and shipping policies
FR-021	The system shall allow dealers to reset their password via email

5.1.4 Admin Login

ID	Requirement
FR-022	The system shall allow administrators to login through a secure admin panel
FR-023	The system shall support multi-factor authentication (MFA) for admin accounts
FR-024	The system shall maintain a log of all admin login activities

5.2 Dealer Management Module

ID	Requirement
FR-025	The system shall allow dealers to create and customize their business profile page including logo, banner, description, and contact details
FR-026	The system shall display a dealer's country, business type, years of operation, and verified badge on their profile
FR-027	The system shall allow dealers to set their store's operating hours and response time
FR-028	The system shall allow dealers to define their accepted payment methods and shipping policies
FR-029	The system shall display dealer ratings and customer reviews on the dealer profile page
FR-030	The system shall allow dealers to view their store performance metrics including total sales, views, and ratings
FR-031	The system shall allow dealers to temporarily pause their store (vacation mode)
FR-032	The system shall allow dealers to manage multiple staff accounts under one dealer profile with role-based permissions

5.3 Product & Inventory Management Module

ID	Requirement
FR-033	The system shall allow dealers to add new spare parts listings with the following details: part name, part number, brand, condition (new/used/refurbished), price, quantity, description, and images
FR-034	The system shall allow dealers to specify vehicle compatibility for each part including make, model, year, and engine type
FR-035	The system shall allow dealers to categorize products under predefined categories and subcategories
FR-036	The system shall allow dealers to upload multiple images (minimum 5) for each product listing
FR-037	The system shall allow dealers to set product pricing in their local currency
FR-038	The system shall allow dealers to apply discounts and promotional pricing on individual products or entire inventory
FR-039	The system shall allow dealers to update stock quantity for each product
FR-040	The system shall automatically mark a product as **Out of Stock** when quantity reaches zero
FR-041	The system shall send a low stock alert notification to the dealer when stock falls below a defined threshold
FR-042	The system shall allow dealers to duplicate existing listings to speed up product entry
FR-043	The system shall allow dealers to activate, deactivate, or delete product listings
FR-044	The system shall allow dealers to import product listings in bulk via CSV or Excel file upload
FR-045	The system shall allow dealers to manage product listings through a centralized inventory dashboard

5.4 Search & Discovery Module

ID	Requirement

FR-046	The system shall provide a search bar allowing users to search spare parts by part name, part number, brand, or vehicle compatibility
FR-047	The system shall provide autocomplete suggestions as the user types in the search bar
FR-048	The system shall display search results ranked by relevance, rating, and availability
FR-049	The system shall allow users to filter search results by the following criteria: category, brand, condition, price range, dealer country, shipping destination, and availability
FR-050	The system shall allow users to sort search results by price (low to high / high to low), rating, newest, and popularity
FR-051	The system shall allow users to search for spare parts by selecting vehicle make, model, year, and engine type
FR-052	The system shall display related products and alternative parts on each product detail page
FR-053	The system shall allow users to compare up to 3 products side by side
FR-054	The system shall display featured and promoted product listings on the homepage and category pages
FR-055	The system shall allow customers to save products to a wishlist for future reference

5.5 Order Management Module

ID	Requirement
FR-056	The system shall allow customers to add products from multiple dealers into a single shopping cart
FR-057	The system shall separate orders by dealer during checkout so each dealer receives their respective order
FR-058	The system shall allow customers to specify quantity for each item in the cart
FR-059	The system shall display a full order summary including product details, quantities, shipping cost, taxes, and total amount before final confirmation
FR-060	The system shall allow customers to apply discount or promo codes during checkout
FR-061	The system shall generate a unique order ID for every placed order

FR-062	The system shall send an order confirmation email to both the customer and the dealer upon successful order placement
FR-063	The system shall allow dealers to confirm, process, or reject incoming orders
FR-064	The system shall allow dealers to update order status at each stage: Confirmed, Packed, Shipped, Out for Delivery, Delivered
FR-065	The system shall allow customers to track their order status in real time from their account dashboard
FR-066	The system shall allow customers to cancel an order before it is shipped
FR-067	The system shall allow customers to submit a return or refund request for delivered orders within a defined return window
FR-068	The system shall automatically generate and send an invoice to the customer upon order confirmation
FR-069	The system shall maintain a complete order history for both customers and dealers

5.7 Shipping & Logistics Module

ID	Requirement
FR-081	The system shall allow dealers to define their shipping zones and supported destination countries
FR-082	The system shall allow dealers to set shipping rates based on weight, dimensions, and destination
FR-83	The system shall display estimated shipping costs to customers during checkout
FR-084	The system shall provide customers with multiple shipping options (standard, express, economy) where available
FR-085	The system shall integrate with third-party logistics providers (DHL, FedEx, ShipStation) for shipment booking and tracking
FR-086	The system shall allow dealers to generate shipping labels directly from the platform
FR-087	The system shall provide customers with a tracking number and live shipment tracking within their order details
FR-088	The system shall send automated shipping notifications to customers at each delivery milestone

FR-089	The system shall support customs declaration information input for international shipments
FR-090	The system shall calculate estimated delivery dates based on shipping method and destination country

5.8 Communication Module

ID	Requirement
FR-091	The system shall provide an in-platform messaging system allowing customers to contact dealers directly
FR-092	The system shall allow customers to send product inquiry or quotation requests to dealers
FR-093	The system shall notify dealers of new messages via email and in-platform notifications
FR-094	The system shall allow dealers to respond to customer inquiries through the messaging system
FR-095	The system shall send automated email notifications to customers for order confirmation, shipping updates, and delivery
FR-096	The system shall send automated email notifications to dealers for new orders, payment receipts, and customer messages
FR-097	The system shall support SMS notifications for critical order and payment updates
FR-098	The system shall maintain a complete message history between dealers and customers

5.9 Review & Rating Module

ID	Requirement
FR-099	The system shall allow customers to submit a star rating (1–5) and written review for a dealer after order delivery
FR-100	The system shall allow customers to submit a star rating and review for individual products after purchase
FR-101	The system shall display the average rating and total number of reviews on dealer profiles and product pages
FR-102	The system shall allow dealers to publicly respond to customer reviews

FR-103	The system shall allow customers to report inappropriate or fake reviews
FR-104	The system shall allow admin to moderate, approve, or remove reviews flagged as inappropriate
FR-105	The system shall prevent customers from submitting multiple reviews for the same order

5.10 Admin Control Module

ID	***Requirement***
FR-106	The system shall provide admins with a dashboard displaying platform statistics including total dealers, customers, orders, and revenue
FR-107	The system shall allow admins to review, approve, or reject dealer registration applications
FR-108	The system shall allow admins to suspend or permanently ban dealer or customer accounts
FR-109	The system shall allow admins to review and moderate product listings for quality and compliance
FR-110	The system shall allow admins to manage product categories and subcategories
FR-111	The system shall allow admins to configure platform commission rates per transaction
FR-112	The system shall allow admins to manage supported countries, currencies, and shipping zones
FR-113	The system shall allow admins to view and manage all transactions and payment records
FR-114	The system shall allow admins to handle disputes between dealers and customers
FR-115	The system shall allow admins to generate and export platform reports (sales, users, revenue) in PDF or Excel format
FR-116	The system shall allow admins to send platform-wide announcements to all users
FR-117	The system shall maintain a complete audit log of all admin actions

5.11 Notification Module

ID	Requirement
FR-118	The system shall send real-time in-platform notifications to users for all major events
FR-119	The system shall allow users to configure their notification preferences (email, SMS, in-platform)
FR-120	The system shall send reminders to dealers for pending orders that have not been processed within 24 hours
FR-121	The system shall send promotional and offer notifications to customers based on their browsing history
FR-122	The system shall notify customers when a wishlisted product's price drops or comes back in stock

5.12 Reporting & Analytics Module

ID	Requirement
FR-123	The system shall provide dealers with a sales analytics dashboard showing revenue, orders, and top-selling products
FR-124	The system shall provide dealers with customer demographic insights including top ordering countries
FR-125	The system shall allow dealers to generate and download sales reports for a selected date range
FR 126	The system shall provide admin with platform-wide revenue, commission, and transaction reports
FR-127	The system shall track and display product page views and conversion rates for dealers

6. Interface Requirements

6.1 User Interface (UI)

The Spare Grid shall provide a clean, responsive, and intuitive web-based interface accessible through all modern browsers. The UI shall include:

- A **homepage** displaying featured products, top dealers, and search functionality
- A **dealer dashboard** for managing products, orders, earnings, and store settings
- A **customer portal** for browsing, ordering, tracking, and account management
- An **admin panel** for platform management, dealer approvals, and reporting
- All interfaces shall be fully responsive and compatible with desktop, tablet, and mobile screen sizes

6.2 Hardware Interface

The Spare Grid does not require any dedicated hardware. It shall operate on:

- Standard web servers hosted on cloud infrastructure (AWS, Google Cloud, or Azure)
- Any client device (PC, laptop, tablet, smartphone) with a modern web browser and internet connection
- No special hardware peripherals are required from the user side

6.3 Software Interfaces

The Spare Grid shall communicate with the following external software systems:

External System	Purpose	Communication Method
Stripe / PayPal API	Payment processing and refund handling	REST API over HTTPS
Currency Exchange API (e.g., Open Exchange Rates)	Real-time currency conversion for multi-currency display	REST API / JSON response
DHL / FedEx / ShipStation API	Shipping rate calculation, label generation, and shipment tracking	REST API over HTTPS
SendGrid / Twilio API	Email and SMS notification delivery	REST API / SMTP
Google Maps API	Dealer location display and regional filtering	REST API / JavaScript SDK

OAuth Providers (Google, Facebook)	Social login for customers	OAuth 2.0 Protocol
Cloud Storage (e.g., AWS S3)	Storage of product images and dealer documents	REST API
Database Server (MySQL / PostgreSQL)	Storing and retrieving all platform data	SQL queries over internal network

6.4 Communication Interface

- All data exchanged between client and server shall be transmitted over **HTTPS using SSL/TLS encryption**
- The platform shall use **RESTful APIs** for communication between frontend and backend
- Data shall be exchanged in **JSON format** between internal services and third-party APIs
- Real-time order and notification updates shall be delivered using **WebSockets or Server-Sent Events (SSE)**
- Email communications shall follow **SMTP protocol** through integrated email service providers

7. Performance Requirements

7.1 Static Performance Requirements

Static requirements define the capacity the system must support without constraining execution behavior:

Requirement	Specification
Concurrent Users	The system shall support a minimum of 10,000 concurrent users without performance degradation
Registered Dealers	The system shall support up to 50,000 registered dealer accounts
Registered Customers	The system shall support up to 1,000,000 registered customer accounts
Product Listings	The system shall handle up to 5,000,000 active product listings across all dealers

Simultaneous Transactions	The system shall support up to 1,000 simultaneous payment transactions
Data Storage	The system shall provide scalable storage to accommodate growing product images, documents, and records

7.2 Dynamic Performance Requirements

Dynamic requirements define constraints on system execution behavior and response time:

Requirement	Specification
Page Load Time	All pages shall load within **3 seconds** under normal network conditions
Search Response Time	Search results shall be returned within **2 seconds** of query submission
Payment Processing Time	Payment transactions shall be processed and confirmed within **5 seconds**
Order Placement Time	Order confirmation shall be generated within **3 seconds** of successful payment
API Response Time	All internal API calls shall respond within **1 second** under normal load
Notification Delivery	Email notifications shall be delivered within **1 minute** of the triggering event
System Uptime	The platform shall maintain a minimum uptime of **99.9%** (approximately 8.7 hours downtime per year)
Maximum Error Rate	The system shall maintain a maximum acceptable error rate of **0.1%** of all transactions
Database Query Time	All database queries shall execute within **500 milliseconds** under normal load
File Upload Speed	Product image uploads shall be processed within **10 seconds** for files up to 10MB

7.3 Scalability Performance

- The system shall be designed to **horizontally scale** by adding additional server instances during peak traffic periods

- The database shall support **read replicas** to distribute query load during high traffic
- A **Content Delivery Network (CDN)** shall be used to deliver static assets (images, CSS, JS) faster to global users

8. Design Constraints

8.1 Technology Constraints

The development team shall adhere to the following technology decisions:

Constraint	Details
Frontend Framework	Must be developed using a modern framework such as React.js or Vue.js for dynamic UI rendering
Backend Framework	Must use Laravel (PHP), Node.js, or Django (Python) for server-side logic
Database	Must use a relational database (MySQL or PostgreSQL) for structured data storage
Cloud Hosting	Must be deployed on a recognized cloud provider (AWS, Google Cloud, or Azure)
Version Control	All source code must be managed using Git with a defined branching strategy

8.2 Security Constraints

- All passwords must be stored using **bcrypt hashing** and never in plain text
- All financial transactions must comply with **PCI-DSS (Payment Card Industry Data Security Standard)**
- The platform must implement **SSL/TLS encryption** for all data transmissions
- The system must enforce **role-based access control (RBAC)** to restrict unauthorized access
- All API endpoints must be protected using **authentication tokens (JWT or OAuth 2.0)**

8.3 Regulatory & Legal Constraints

- The platform must comply with **GDPR (General Data Protection Regulation)** for users in the European Union

- The platform must provide a clear **Privacy Policy and Terms of Service** agreement during registration
- The system must support **VATMOSS** or equivalent tax regulations for international digital transactions
- Dealer registration must comply with **Know Your Business (KYB)** verification standards

8.4 Resource & Hardware Constraints

- The platform must be optimized to function on devices with **minimum 2GB RAM** and standard internet speeds
- The system must be functional on browsers with **JavaScript enabled**; it shall not rely on deprecated browser technologies
- All uploaded files shall have a **maximum size limit of 10MB per file** to manage server storage efficiently

8.5 Operational Constraints

- The system must provide a **maintenance mode** that displays a user-friendly message during scheduled downtime
- Database **backups must be performed daily** and stored securely for a minimum of 30 days
- The platform must log all **critical system errors** for debugging and monitoring purposes

9. Non-Functional Attributes

9.1 Security

- The Spare Grid shall implement end-to-end encryption for all sensitive data including personal information and payment details
- Multi-factor authentication shall be available for dealer and admin accounts
- The system shall conduct regular security audits and vulnerability assessments
- Protection against common web threats such as SQL Injection, XSS, and CSRF attacks shall be enforced

9.2 Reliability

- The system shall achieve a minimum uptime of **99.9%** ensuring continuous availability to global users across all time zones
- Automated failover mechanisms shall be in place to maintain service in case of server failure
- The system shall perform daily automated backups to prevent data loss

9.3 Portability

- The Spare Grid shall be accessible on all major operating systems including Windows, macOS, Linux, iOS, and Android through standard web browsers
- The platform shall not require any software installation on the client side
- The codebase shall be structured to allow future migration to different hosting providers without major re-engineering

9.4 Scalability

- The system architecture shall support horizontal scaling to accommodate growing numbers of dealers, customers, and product listings
- The database shall be designed to handle millions of records without significant performance degradation
- Cloud auto-scaling shall be configured to handle sudden spikes in traffic automatically

9.5 Maintainability

- The codebase shall follow clean coding standards and be fully documented to allow easy maintenance and future enhancements
- The system shall use modular architecture so individual components can be updated without affecting the entire platform
- A staging environment shall be maintained for testing updates before deploying to production

9.6 Usability

- The platform shall provide an intuitive and user-friendly interface requiring no technical expertise from dealers or customers

- The system shall display meaningful error messages guiding users to correct actions
- The platform shall support keyboard navigation and basic accessibility standards (WCAG 2.1)

9.7 Data Integrity

- The system shall validate all user inputs to prevent incorrect or malicious data from entering the database
- All financial transactions shall be atomic, ensuring no partial or duplicate payments occur
- The system shall maintain consistent data across all modules using database transactions and constraints

9.8 Reusability

- The system shall be built using reusable components and modular APIs that can be extended for future features such as a mobile application
- Common services such as notifications, authentication, and payment processing shall be built as independent reusable modules

9.9 Application Compatibility

- The Spare Grid shall be fully compatible with the latest versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari
- The platform shall maintain basic functionality on browsers that are one major version behind the latest release

10. Preliminary Schedulethe

10.1 Preliminary Project Schedule

The development of The Spare Grid is estimated to be completed in **10 months** divided into the following phases:

Phase	Activities	Duration

Phase 1 – Requirements & Planning	Finalize SRS, project planning, team setup, technology selection	3 Weeks
Phase 2 – UI/UX Design	Wireframing, prototyping, design approval for all modules	4 Weeks
Phase 3 – Database Design	Entity-relationship diagram, schema design, database setup	2 Weeks
Phase 4 – Backend Development	Core modules: auth, dealer, product, order, payment, shipping	4 Weeks
Phase 5 – Frontend Development	Implementing UI for customer portal, dealer dashboard, admin panel	6 Weeks
Phase 6 – Testing & QA	Unit testing, integration testing, performance testing, UAT	5 Weeks
Phase 7 – Deployment & Launch	Server setup, deployment, go-live preparation, final review	2 Weeks
Phase 8 – Post-Launch Support	Bug fixes, performance monitoring, minor enhancements	4 Weeks

Total Estimated Duration: 46 Weeks (approximately 10–11 months)

11. Appendices

11.1 References

Reference	Source
IEEE Std 830-1998	IEEE Recommended Practice for Software Requirements Specifications
GDPR Official Text	https://gdpr.eu
PCI-DSS Standards	https://www.pcisecuritystandards.org
Stripe API Documentation	https://stripe.com/docs/api
PayPal Developer Documentation	https://developer.paypal.com
DHL API Documentation	https://developer.dhl.com
Open Exchange Rates API	https://openexchangerates.org
Google Maps Platform	https://developers.google.com/maps
WCAG 2.1 Accessibility Guidelines	https://www.w3.org/TR/WCAG21

11.2 Definitions

Term	Definition
Dealer	A registered business or individual selling motor spare parts on The Spare Grid platform
Customer	A registered user who browses and purchases spare parts through the platform
Admin	An internal platform operator responsible for managing and monitoring all platform activities
Listing	A product entry created by a dealer showcasing a spare part for sale
Order	A confirmed purchase request made by a customer for one or more spare parts
Escrow	A financial arrangement where the platform holds payment until order delivery is confirmed
Commission	A percentage fee deducted by the platform from each dealer transaction
KYB	Know Your Business – a verification process to confirm the legitimacy of registered dealers
SKU	Stock Keeping Unit – a unique identifier assigned to each product listing
Wishlist	A saved list of products a customer is interested in purchasing in the future
Vacation Mode	A feature allowing dealers to temporarily pause their store without deleting their account

11.3 Acronyms and Abbreviations

Acronym	Full Form
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience
API	Application Programming Interface
REST	Representational State Transfer
HTTPS	Hypertext Transfer Protocol Secure
SSL	Secure Sockets Layer
TLS	Transport Layer Security
JWT	JSON Web Token

GDPR	General Data Protection Regulation
PCI-DSS	Payment Card Industry Data Security Standard
CDN	Content Delivery Network
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
MFA	Multi-Factor Authentication
WCAG	Web Content Accessibility Guidelines
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
SQL	Structured Query Language
AWS	Amazon Web Services

12. Uses of SRS Document

The SRS document for **The Spare Grid** serves the following purposes for different stakeholders:

- **Development Team** uses it as a technical blueprint to build each module of the platform exactly according to the defined requirements, ensuring nothing is missed or misunderstood
- **Testing & QA Team** uses it to design test cases and validate that every functional and non-functional requirement has been correctly implemented in the final product
- **Maintenance & Support Staff** refer to it to understand the intended behavior of the system when diagnosing bugs or planning future updates
- **Project Manager** uses it to plan the project timeline, allocate resources, estimate effort, and track progress against defined requirements
- **Clients & Stakeholders** rely on it to understand exactly what the platform will deliver, setting clear and realistic expectations before development begins
- **As a Legal Contract** it serves as a formal agreement between the development team and the client defining the scope, features, and boundaries of the project
- **Documentation Purpose** it acts as a permanent reference document for the entire project lifecycle including future enhancements and scaling decisions

13. Conclusion

The development of **The Spare Grid** requires this well-structured Software Requirements Specification as its foundation. This SRS document has comprehensively defined the purpose, scope, functional requirements, interface requirements, performance benchmarks, design constraints, and non-functional attributes of the platform.

By serving as a single source of truth, this document ensures that all stakeholders including dealers, customers, developers, testers, and project managers share a unified understanding of what The Spare Grid is designed to achieve. It provides the development team with a clear roadmap, guides the QA team in validating the system, informs project management decisions regarding schedule and budget, and sets transparent expectations for the client.

Adherence to this SRS will ensure that **The Spare Grid** is delivered as a secure, scalable, high-performing, and user-friendly global marketplace for motor spare parts — on time, within budget, and to the highest quality standards.